PLANNING & ESTIMATION IN IT PROJECT MANAGEMENT

1. INTRODUCTION TO PLANNING

Planning transforms an idea into an executable roadmap.

Objectives:

- Define scope
- Organize tasks
- Estimate time and cost
- Allocate resources
- Reduce uncertainty

Key principle:

You cannot control what you have not planned.

2. PROJECT SCOPE DEFINITION

2.1 Formal Definition

The scope defines the boundaries of the project and ensures alignment between stakeholders.

2.2 Scope Elements

A. Functional Requirements : What the system does

Example:

- User login
- Data storage

B. Non-functional Requirements :How the system performs

Example:

- Response time < 2 seconds
- Secure authentication

C. Constraints

Budget

Deadline

Technology

D. Assumptions

Internet availability

Skilled developers

2.3 Real Example

Project: University ERP Module

Included:

- Student registration
- Grade management
- Reporting

Excluded:

- Mobile app
- AI analytics

2.4 Scope Validation

Tools:

- Stakeholder meetings
- Requirement documents
- Prototypes

2.5 Scope Risks

Risk	Impact
Unclear requirements	Rework
Scope creep	Delay
Miscommunication	Failure

3. WORK BREAKDOWN STRUCTURE (WBS)

3.1 Definition

WBS = hierarchical decomposition of the project.

3.2 Rules (Very Important)

- Each task must be measurable

- Tasks must be independent
- No overlap
- Clear deliverables

3.3 Example (Detailed)

Project: E-learning Platform

Level 1:

Project definition

Level 2:

Analysis

Design

Development

Testing

Level 3 (Development):

Backend

Frontend

Database

Level 4:

API endpoints

UI components

SQL schema

3.4 WBS Benefits

- Accurate estimation
- Clear responsibilities
- Risk identification
- Better monitoring

4. PROJECT SCHEDULING

4.1 GANTT CHART

Definition

Timeline representation of tasks.

Example

Task	Duration	Start	End
Analysis	2w	Week 1	Week 2
Design	2w	Week 3	Week 4
Dev	6w	Week 5	Week 10

Key Concepts

Dependencies

Parallel tasks

Milestones

4.2 PERT / CPM

Concepts

Nodes = tasks

Arrows = dependencies

Example

Tasks:

A → B → D

A → C → D

Critical Path

Longest path = determines project duration

Importance

Focus on critical tasks

Optimize scheduling

5. ESTIMATION

5.1 TIME ESTIMATION

Methods

- Expert Judgment : Based on experience
- Analogous : Compare with similar projects
- Parametric : Mathematical models

PERT Formula

T = (O + 4M + P) / 6

Example

O = 3 days

M = 5 days

P = 9 days

T = (3 + 20 + 9)/6 = 5.3 days

5.2 RESOURCE ESTIMATION

Types

- Human
- Technical
- Financial

Example :

Team :

1 Project Manager

2 Backend dev

2 Frontend dev

1 QA

5.3 COST ESTIMATION

Formula

Cost = Σ (Resource × Duration)

Example

Developer:

150,000 DA × 6 months × 3 dev = 2,700,000 DA

6. COCOMO MODEL

Types :

- Basic
- Intermediate
- Detailed

Formula :

Effort = a × (KLOC)^b

Example

KLOC = 20

Effort ≈ 50 person-months

Interpretation

Helps define team size

Helps budget estimation

7. INTEGRATED PROCESS

Steps:

1) Define scope
2) Build WBS
3) Estimate time
4) Assign resources
5) Estimate cost
6) Build schedule

8. COMMON MISTAKES

- Underestimating complexity
- Ignoring risks
- Poor communication
- Weak WBS

9. BEST PRACTICES

- Iterative planning
- Team involvement
- Continuous validation
- Buffer inclusion

PRACTICAL LAB— STUDENT PROJECT

Objective

Students must plan and estimate a their project from scratch.

TASK 1 — DEFINE SCOPE

Students must :

- Define included features
- Define excluded features
- Identify constraints

TASK 2 — BUILD WBS

Students must create WBS diagram or table :

- Minimum 3 levels
- At least 25 tasks

TASK 3 — GANTT PLANNING

- Define duration of each task
- Identify dependencies

Deliverable: Gantt chart

TASK 4 — PERT ANALYSIS

- Draw network diagram
- Identify critical path

Deliverable: PERT diagram

TASK 5 — ESTIMATION

- **Time**: Use PERT formula
- **Resources**: Define team
- **Cost**: Calculate total budget

Deliverable: Estimation report

TASK 6 — COCOMO

- Estimate KLOC

- Calculate effort

Deliverable: Effort estimation

BONUS (ADVANCED)

- Add risk analysis
- Add Agile version (Scrum plan)
- Compare methods